

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

20ECE84 – OBJECT ORIENTED PROGRAMMING WITH C++

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

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|---|--|-----|----|---|---|
| 1. a) Explain the different types of loops available in C++ code by taking an example of generating first 'n' natural numbers. | 10 | L*3 | 1 | 1 | |
| b) Why C++ is called an Object-oriented programming language. Explain with sample code. | 10 | L2 | 2 | 1 | |
| 2. a) What is a pointer? Explain pointers concerning functions with sample C++ code written to read the name and age of a user. | 05 | L3 | 2 | 1 | |
| b) Write the complete output for the C++ code given below: | | | | | |
| <pre>#include <iostream> using namespace std; void guess_me(int& abc, int xyz, int& pqr); int main() { int a, b, c; a = 1; b = 2; c = 3; guess_me(a, b, c); cout << a << " " << b << " " << c; return 0; }</pre> | <pre>void guess_me(int& abc, int xyz, int& pqr) { cout << abc << " " << xyz << " " << pqr << endl; abc = 0; xyz = -1; pqr = -2; cout << abc << " " << xyz << " " << pqr << endl; }</pre> | 05 | L2 | 2 | 1 |
| c) What is Call by value method? What will the output be, when '&' is completely removed from the above code? | 05 | L2 | 2 | 1 | |
| d) Considering the case of finding the area of a rectangle (inputs width and height, output area of rectangle), demonstrate the Function and Call by value technique. (use functions read() and disparea()) | 05 | L3 | 2 | 1 | |
| 3. a) Define Constructor and Destructor. Explain different types of constructors available in C++ by taking the example of a website where the constructor is used to store default values like users name, age and mobile number. Also, in the same code show the use of a destructor. (Note: Name the class as 'nittesite') | 10 | L3 | 2 | 1 | |
| b) What is a Friend function? Explain. Also, how a friend function can help in determining whether the member defined in one class is equal to the member defined in the other class. | 10 | L3 | 2 | 1 | |

Unit – II

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|-------|---|----|----|---|---|
| 4. a) | What are the different options provided in C++ to manage a file? How to write to a file and read from a file named "nitte.dat". Explain with a sample C++ code. | 10 | L3 | 4 | 1 |
| b) | What is Polymorphism? Explain its types by taking a sample code for each one. | 10 | L2 | 3 | 1 |
| 5. a) | Briefly explain the input and output stream classes provided by C++ and how they can perform different I/O operations. | 10 | L2 | 4 | 1 |
| b) | Write a C++ program to create a Car class. The class should contain the features of Cars like Name, Colour, Price and CC as members. Include the member functions in the Car class, to read and show the values of Car class members separately (a member function to store and display each member individually). Also, derive class Maruti of type Car. Demonstrate how to store and display members of class Maruti. | 10 | L3 | 3 | 1 |
| 6. a) | What is Inheritance? Explain different types of Inheritance, the role of access specifiers and their importance in inheritance. | 10 | L2 | 3 | 1 |
| b) | With the help of Sample C++ code, explain the Virtual and Pure virtual functions. | 10 | L2 | 3 | 1 |

Unit – III

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|-------|---|----|----|-----|---|
| 7. a) | What is a Class template? Explain with the help of a sample C++ code. | 10 | L2 | 5 | 1 |
| b) | What is an Exception? Explain the use and syntax of Multiple exceptions by taking a case that generates exceptions of type string, integer and char when the entered integer 'ch' is 0, 1 and -1. | 10 | L3 | 5 | 1 |
| 8. a) | With respect to object-oriented programming language, explain function template by taking a sample C++ code. | 10 | L2 | 1,5 | 1 |
| b) | What is the role played by the Standard template library in C++? Also, explain its components. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
